

PROJECT 33

Design and Implementation of a Database Management System for Training Programme and Certification Management

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Problem Identification

Why training centres need a better system.

The Situation Today

Training centres run many courses, batches and trainers at once. They must track enrolment, attendance, scores, fees and certificates for every student.

Most of this is still done manually — on paper or in scattered spreadsheets that don't talk to each other.



Wrong certificate eligibility

Students can receive a certificate even without enough attendance or a passing score.



Duplicate enrolments

The same learner can accidentally get enrolled twice in one batch.



Slow, error-prone reports

Attendance, marks and fee data live in separate files, so reports take long and often have mistakes.

Project Scope

What this project will and will not cover.

In Scope

Course, batch, trainer and learner records

Enrolment, session and attendance tracking

Assessments, scores and fee records

Certificate issuing and renewal tracking

SQL reports: batch lists, attendance, performance, fees

A simple Java/Python app for data entry

Out of Scope

Online payment gateway integration

Native mobile app version

Multi-branch / multi-city support

E-learning video / content delivery

Project Objectives

What we want to achieve.

1

Find out what the training centre really needs (requirements).

2

Draw a clear diagram (ER diagram) of all the data and how it connects.

3

Turn that diagram into real database tables, organised correctly (up to 3NF).

4

Add rules so wrong or duplicate data cannot be saved.

5

Write SQL queries and reports (attendance, fees, results, certificates).

6

Build a simple front-end app so staff can use the system easily.

Who Will Use the System?

Four types of users, each with a different role.



Administrator

Sets up courses, batches and trainers. Manages the whole system.



Trainer

Takes attendance and enters scores for their sessions.



Student / Learner

Enrols in courses and checks their attendance, fees and certificate.



Office Staff

Handles fee payments and prints certificates.

Functional Requirements

What the system must be able to do.

1 Admin can add and update courses, batches, and assign trainers to each batch.

2 System stops a batch from taking more learners than its seat limit.

3 Trainers can mark attendance and enter scores for every session.

4 System blocks a learner from enrolling twice in the same batch.

5 Learners can view their own attendance, scores, fees and certificate status.

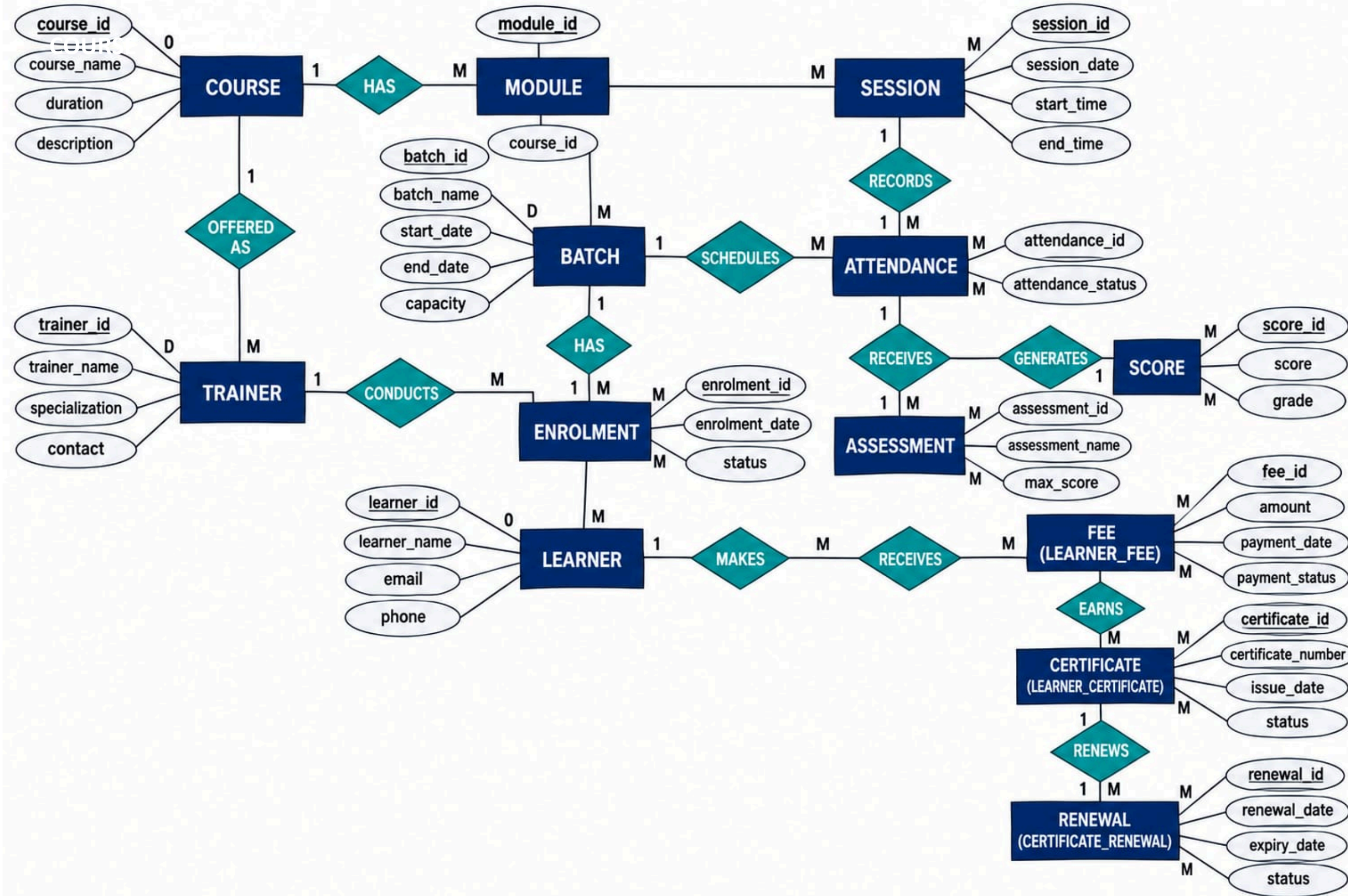
6 System checks attendance % and score before allowing a certificate to be issued.

7 Office staff can record fee payments and issue certificates with a unique number.

8 System can generate reports: batch lists, attendance, performance, fees and renewals.

Conceptual Design — ER Diagram

Lines show how the entities connect to each other.



How the Entities Are Connected

The ER diagram in plain words — every link is one-to-many (1:M).

A Course *has many* Modules and Batches

A Trainer *is assigned to* many Batches

A Learner *can have many* Enrolments

A Batch *has many* Enrolments and Sessions

A Session *belongs to* one Batch and one Module

An Enrolment *leads to many* Attendance records

A Batch *has many* Assessments

An Assessment *produces many* Scores

An Enrolment *produces one* Fee record

An Enrolment *can lead to one* Certificate

A Certificate *may have many* Renewals

Initial Relational Schema

Proposed tables, primary keys and how they relate.

Table	Primary Key	Foreign Key(s)	Related To
Course	course_id	—	—
Module	module_id	course_id	Course
Trainer	trainer_id	—	—
Batch	batch_id	course_id, trainer_id	Course, Trainer
Learner	learner_id	—	—
Enrolment	enrolment_id	learner_id, batch_id	Learner, Batch
Session	session_id	batch_id, module_id	Batch, Module
Attendance	attendance_id	session_id, enrolment_id	Session, Enrolment
Assessment	assessment_id	batch_id, module_id	Batch, Module
Score	score_id	assessment_id, enrolment_id	Assessment, Enrolment
Fee	fee_id	enrolment_id	Enrolment
Certificate	certificate_id	enrolment_id	Enrolment
Renewal	renewal_id	certificate_id	Certificate

Summary — Ready for Review 1

What has been prepared so far.

1

Problem, scope, objectives, users and functional requirements are documented.

2

The ER diagram shows 13 entities and how they connect.

3

An initial relational schema proposes tables, primary keys and relationships.

Next Step

After Review 1, we will build the actual database, add sample data, write SQL queries and reports, and connect the front-end app.